

EEG-Based Analysis of Alzheimer's Disease

Related Works Table:

<i>Title</i>	<i>Author s</i>	<i>Ye ar</i>	<i>Dataset</i>	<i>Applicati on Area</i>	<i>Method/Mo del</i>	<i>Metrics</i>	<i>Strengths</i>	<i>Limitatio ns</i>	<i>Research Gap</i>	<i>Relation to Work</i>	<i>Type of GNN</i>
<i>Graph neural networks for electroencephalogram analysis: Alzheimer's disease and epilepsy use cases</i>	Sergi Abadal et al.	2025	Private AD dataset (63 AD, 63 MCI, 63 HC)	AD diagnosis and seizure discrimination	Graph Transformer architecture	Ternary AD Accuracy: 89.02% ; Seizure Accuracy: 96.95%	Captures non-Euclidean relations; eliminates the need for pre-defined edge weights; high explainability	Higher number of trainable weights compared to vanilla GCN	Prior works were limited to simple GNNs or binary classification; they lacked large-scale design explorations	Directly validates the use of Graph Transformers (attention-based GNNs) for AD detection	Graph Transformer
<i>Variational Mixture of Graph Neural Experts for Alzheimer's Disease Recognition across Frequency Bands</i>	Jun-En Ding et al.	2025	Open AD (88 subjects) and Session-based AD (123 subjects)	AD/FTD subtyping and dementia staging	Variational Mixture of Graph Neural Experts (VMoGE)	AUC for HC vs AD: 0.89	Models' latent uncertainty; uses expert specialized frequency bands; provides neurophysiological interpretable markers	Single-modality limitation; AD neurodegeneration is multi-level and complex	Prior methods were hindered by full-band frequency analysis and lacked comprehensive evaluations with clinical indicators	Implements a variational gating mechanism to integrate multi-band EEG features	Variational GNN / Mixture-of-Experts

<i>Adaptive Gated Graph Convolutional Network for Explainable Diagnosis of Alzheimer's Disease Using EEG Data</i>	Domini k Klepl et al.	2023	20 AD patients and 20 Healthy Controls (Sheffield memory clinic)	Explainabl e AD diagnosis	Adaptive Gated Graph Convolution al Network (AGGCN)	Accuracy (Eyes-Closed): 89.1% ; AUC: 0.895	Adaptively learns graph structures; gating filters info across spatial scales; explainable predictions via ASAP pooling	Small sample size; short window lengths may lose low-frequency signal info	GNN-based AD diagnosis remains relatively unexplored; prior studies used simple GNNs without gating	The gating mechanism adaptively weights spatial scales aligning with our requirement	Gated Graph Convoluti onal Network
<i>Flexible and Explainable Graph Analysis for EEG-based Alzheimer's Disease Classification</i>	Jing Wang et al.	2023	108 AD patients and 15 Healthy Control (Taipei Veteran General Hospital)	AD classificati on across different dementia stages (CDR 0.5–2+)	Flexible GGCN with MOTPE hyperparam eter tuning	AUC: >0.90 (Moderate to Severe Dementia); high precision/r ecall	Highly flexible; optimizes block numbers for clinical metrics; reveals regional connectivity differences in brain lobes	Specificity variability in mild cases	Need for better interpretability of adjacency matrices to reveal regional connectivity differences	Focuses on flexibility and explainability using gated mechanisms	Gated Graph Convoluti onal Network
<i>Comorbidity-based framework for AD</i>	Ferial Abuhantash et al..	2024	ADNI (1331 subjects) & AIBL	Early AD prediction using EHR data	Neighborhood Patient Graph with ChebConv/ GAT	Multi-class Acc: 0.98 ; AD/CN Acc: 0.99	Innovative use of EHR comorbidity data; cost-effective	Sensitive to feature assignme nt; performance	EHR-based comorbidity data for AD GNN classification	Provides performance benchmarks for GAT (Graph Attention	GAT, ChebConv , GraphSA GE

<i>classification using GNNs</i>			(858 subjects)				alternative to imaging	modulations if comorbidities removed	largely overlooked	Network) in AD classification.	
<i>A novel hybrid model in the diagnosis and classification of Alzheimer's disease: Deep ensemble learning approach</i>	Majid Nour et al.	2024	Combined public datasets from Florida State University (104 AD, 36 HC)	AD and Healthy Control classification using raw EEG	Deep Ensemble Learning (DEL) with five 2D-CNN classifiers	Average Accuracy: 97.9%	Achieves state-of-the-art accuracy; automated feature extraction; robust against overfitting	Lacks interpretability; further effort needed to understand clinical application impacts	Limited previous use of deep ensemble learning in EEG-based AD classification	Uses the exact same public dataset we have chosen; provides the CNN-based baseline we need	N/A (CNN-based Ensemble)